



Human-Centered Explainability for Medical AI with Metaheuristic Optimization and LLM Reporting

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1 Abstract

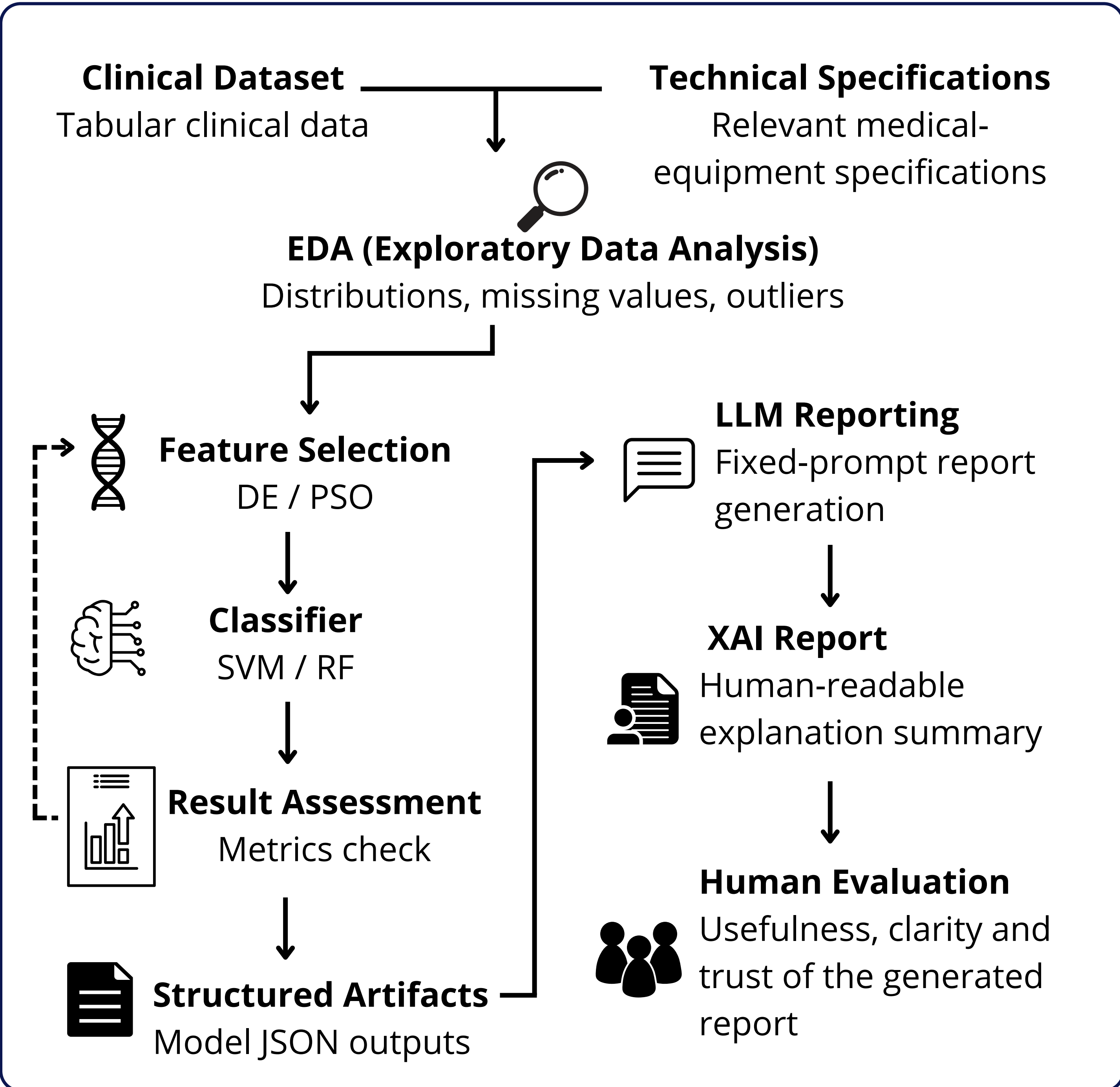
Artificial Intelligence is increasingly embedded in medical equipment, but its internal behavior often remains opaque to healthcare professionals. This paper presents a human-centered XAI pipeline that translates system-level AI behavior into accessible reports using metaheuristic feature selection, supervised classification, structured artifacts, and fixed-prompt LLM reporting. Experiments on three tabular clinical datasets showed competitive performance, compact feature subsets, and dataset-dependent stability patterns. Human evaluation indicated that the reports were clear, coherent, and practically interpretable. These results support reproducible, system-level explainability for improving human understanding of AI-assisted medical equipment.

2 Contribution & Motivation

A human-centered pipeline that combines metaheuristic feature selection, predictive modeling, structured artifacts, and LLM-based reporting for accessible system-level understanding.

- Healthcare professionals may use AI-enabled medical equipment without a clear view of how its embedded AI reaches its results.
- Many XAI approaches focus on instance-level explanations rather than system-level understanding.
- Healthcare stakeholders need transparent summaries of variable relevance, model behavior, and system stability.

3 Proposed Pipeline



Highlight

DE + SVM: balanced solution
Positive human evaluation: 3.87/5

4 Experimental Setup & Results

Datasets: Heart Disease, Diabetes, Breast Cancer

Methods: DE, PSO, SVM, RF

Validation: Stratified 5-fold CV, explicit leakage control

Metrics: Accuracy, Precision, Recall, F1, ROC-AUC

LLM Reporting: GPT-4.1; fixed prompts; no raw patient data

Human Study: Healthcare professionals; adapted from the System Causability Scale (SCS)

Comparative Summary of Method Combinations

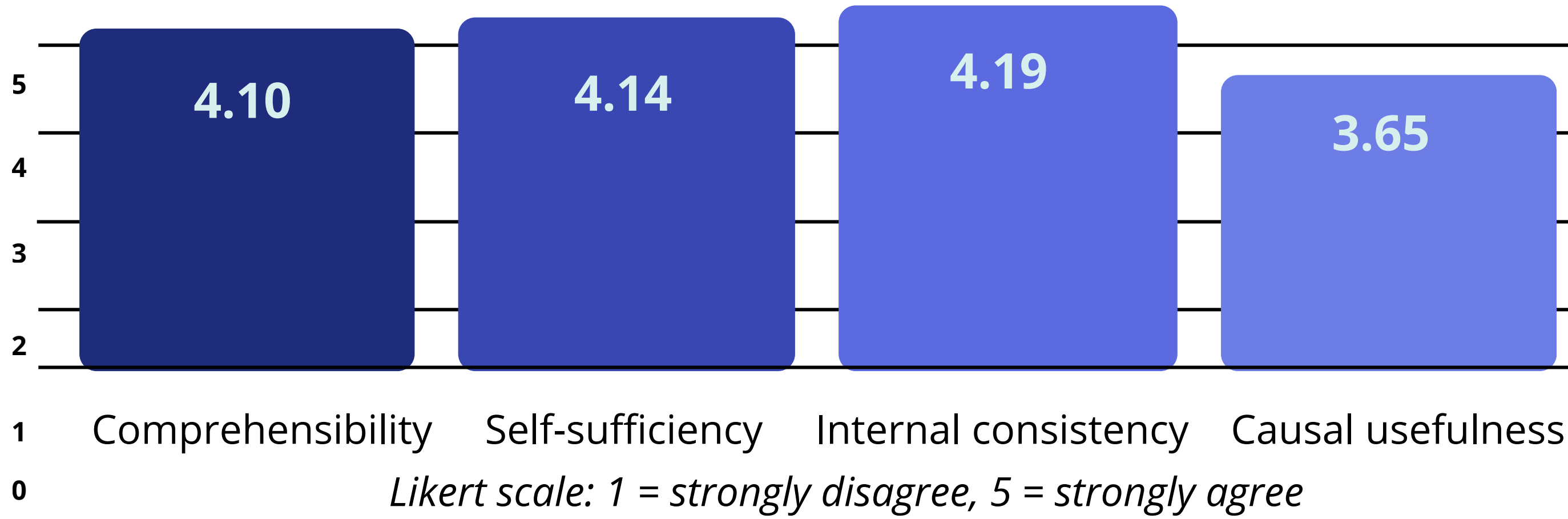
Method	ROC-AUC	Selected Features	Runtime (s)
DE + SVM	0.99	10.8	14.7
DE + RF	0.987	11	574.8
PSO + SVM	0.995	14.8	14.1
PSO + RF	0.992	15.6	569.6

This comparison highlights not only the top-performing model, but also the most balanced methodological configuration.

DE + SVM provided a compact solution with competitive ROC-AUC and low runtime.

Human evaluation

Overall acceptance: 3.87 / 5 (77.38%)



Healthcare specialists positively rated the clarity, consistency, and practical interpretability of the generated XAI reports.

5 Key Findings & Conclusion

- DE + SVM provided a compact solution with competitive ROC-AUC and low runtime.
- Feature stability was dataset-dependent.
- The LLM transformed structured artifacts into accessible XAI reports.
- Healthcare professionals rated the explanations positively.

This work supports the transparent evaluation of embedded AI in medical equipment by combining optimization, predictive modeling, structured artifacts, and human-centered reporting, without replacing clinical judgment or producing patient-level recommendations.

Future work: broader datasets, expanded technical configurations, real-world medical equipment scenarios, and richer human evaluation.



Repository
<https://mauriciopriego.github.io/cbms-2026-xai-supplementary-materials/>

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